

Not every stable cluster is homeostatic: Stability, network order, and control in projectible kinds

Brett Reynolds 

Humber Polytechnic & University of Toronto

28th May 2026

Abstract

Homeostatic property cluster theory gave natural-kinds realism a powerful anti-essentialist form, on which a kind can be real because its properties cluster through mechanisms rather than because it has an essence (Boyd, 1991, 1999). The word HOMEOSTATIC has tended to carry more weight than the theory has earned, because it is used for at least three things that are not the same: reliable co-occurrence, internal causal order, and feedback-sensitive control. This paper treats projectibility as the target of kind claims and offers a diagnostic for distinguishing the ways it can be secured. A stable property cluster requires reliable co-occurrence together with a specified projection. A networked kind requires internal structure among central, derivative, and evidentially useful properties. A control-secured kind requires the more demanding pattern of perturbation-sensitive correction that preserves a higher-scale relation through lower-scale variation. Since the maintenance literature already separates the stabilizers that keep a cluster clustered from the narrower case of a controller, and since Weinberger's account of homeostasis as causal control makes the stronger standard precise (Weinberger, 2026), the distinction can be drawn from within Boyd's framework rather than against it. The Watt governor serves as a calibration case, glycemic regulation as a harder biological extension, English interjections as a negative linguistic case, and agreement monitoring as a candidate positive one. The result is a disciplined way of using Boyd's insight that keeps HOMEOSTATIC from becoming a loose synonym for stable, mechanism-maintained, or projectible.

Keywords: natural kinds; homeostatic property clusters; projectibility; homeostasis; causal control; linguistic categories

I INTRODUCTION

The target of this paper is a quiet ambiguity in recent anti-essentialist kind theory. Boyd's homeostatic property cluster framework, Slater's stable property cluster proposal, and Khalidi's causal-network account all reject essences, and they all aim to explain why some categories support projection while others do not. Each, though, locates the basis for projection somewhere slightly different. Boyd ties

*Contact: brett.reynolds@humber.ca

it to homeostatic mechanisms that maintain the cluster (Boyd, 1991); Slater deliberately sets the stability of a cluster apart from the means by which it is maintained (Slater, 2015); and Khalidi locates it in the causal-network order among a kind's properties (Khalidi, 2013, 2018). The ambiguity lies less in these accounts than in the shared vocabulary of "homeostatic property clusters", which covers all of these options at once and so leaves underdetermined what secures projection in any given case. The looseness is clearest in the homeostatic-mechanism component of the framework. Boyd gives that component considerable flexibility, treating the clustering as "metaphorically (sometimes literally)" homeostatic and allowing it to be sustained by mutual property support, underlying mechanisms, or both (Boyd, 1999). Onishi and Serpico are therefore right that the notion of a homeostatic mechanism in HPC theory has never been fully explicated (Onishi & Serpico, 2022).

The working claim is that not every stable cluster is homeostatic. The literature need not explicitly identify every stable cluster with a homeostatic one for this to matter. Shared HPC vocabulary still makes it too easy to move from stability, maintenance, or network order to the stronger homeostatic label without specifying the securing structure. The correction keeps Boyd's projectibility frame in place while asking that HOMEOSTASIS be used more literally, in a way that makes a genuine demand. Weinberger's recent account of homeostasis as a form of causal control is what supplies that demand (Weinberger, 2026).

The contribution is therefore diagnostic rather than revisionary. It leaves HPC theory in place and separates three explanatory roles that the shared vocabulary tends to compress. Some profiles support projection because their properties recur reliably across the relevant cases. Others support projection because their properties occupy ordered positions in a causal or mixed network, so that some generate or constrain others. A narrower set support projection because a feedback-sensitive organization corrects perturbations and thereby preserves a higher-scale relation. Only this last role gives the word HOMEOSTATIC anything to mean beyond stability.

The scope is Boyd-style projectible realist kind claims. Many classifications earn their keep by mapping variation, stabilizing measurement, organizing historical comparison, or coordinating inquiry before they support strong projection. Those uses are legitimate, but the diagnostic here concerns categories offered as real kinds because they support prediction, explanation, induction, or controlled intervention.

A word about method is in order, since the cases that follow run across a machine, an organism, and two linguistic categories. They are not offered as a survey of those domains. They are chosen because they stress the diagnostic at different points and test whether a single, domain-neutral distinction can tell the three securing structures apart. A distinction in the theory of kinds ought to travel in this way, and if it held only in biology that would be a limitation rather than a sign of rigour. The cases earn their place by what they reveal about the diagnostic, not by representing their fields.

The order of what follows is straightforward. The next four sections develop the target, projectibility, and the three securing structures, ending with the distinction between a stabilizer and a controller. A short section then states the diagnostic as a grid before the remaining sections work through the cases and ask what the distinction changes for HPC theory more broadly.

2 PROJECTIBILITY FIRST

Boyd's framework works because projectibility does not require necessary and sufficient conditions. A category can support induction when its associated properties are connected non-accidentally, even

where no single property is shared by every member. On Boyd's accommodation thesis, successful classification is a matter of fitting our inferential practices to causal structure, so that induction and explanation "always require that we accommodate our categories to the causal structure of the world" (Boyd, 1991, p. 224). We construct the categories, but whether they succeed depends on a structure we do not control, and it is that structure which does the explanatory work.

The problem of PROJECTIBILITY comes from Goodman's new riddle of induction, where a predicate can fit all observed cases and still fail to license projection to unobserved ones (Goodman, 1955/1983). Boyd's contribution was to ground projectibility in causal structure without grounding it in essences, which is also what gave HPC theory its purchase against the worry, which Boyd associated with Hacking, that where a kind's boundaries are fuzzy its membership is fixed by "people in concert" rather than by nature (Boyd, 1991).

This is one reason it helps to begin from projectibility rather than from mechanism. Mechanisms matter to the theory because they explain why projection is reliable, so a category that supports no projection is not redeemed by having a tidy causal story behind it, and a category that does support projection is already doing much of the work a realist about kinds cares about. The task of the rest of the paper is to ask, for categories that are clearly projectible, what kind of structure secures the projection.

The risk is that this question can be made too easy. If the only thing asked is whether a cluster recurs, a great many categories will qualify, because recurrence on its own is cheap. Stable co-occurrence may license some inference, but an analysis still has to specify what is being projected from what before the category becomes theoretically interesting, and that happens only when its profile supports prediction, explanation, or controlled intervention.

3 OBSERVED STABILITY AND INVARIANCE

Slater's stable property cluster account presses the useful point that homeostatic mechanisms are not always needed for natural kindness. What matters on this view is the stability of a property cluster, considered apart from the various ways that stability might be maintained, since a cluster can, in Boyd's phrase, hold itself together (Slater, 2015). This yields the first distinction the paper needs, because a stable and projectible cluster may be perfectly real without being homeostatic in any stronger sense.

Stability that is left unexplained is nonetheless too thin to do all the work, and Onishi and Serpico turn this into a dilemma. If the explanatory role of underlying processes is made optional or postponed, as stable-cluster views invite us to do, the account risks leaving underexplained why some inductions are rational rather than lucky. If causal structure is retained through causal-network order, as Khalidi proposes, the analysis still faces interest-relativity, because which structure counts as the relevant one depends on the explanatory purpose at hand (Onishi & Serpico, 2022). Their broader observation is that the notion of a homeostatic mechanism has never been fully explicated, and that interest-relativity is unlikely to be eliminable.

The diagnostic offered here does not try to remove that interest-relativity so much as to make it explicit. Asking first what a category lets us project, and against which declared expectation, fixes the explanatory target, and the question of securing structure is then asked relative to that target rather than in the abstract. Interest-relativity becomes a parameter the analysis declares rather than a cost it pays without noticing.

Woodward's account of invariance helps locate what mere stability leaves out. The kind of stability that matters for explanation is invariance under intervention, not simply persistence across space and time, so that a generalization earns its explanatory keep when it would continue to hold across a relevant range of changes, including changes that could be imposed (Woodward, 2001). The first diagnostic is correspondingly modest: a stable property cluster should support some projection across a specified range of contexts and interventions, with evidence that goes beyond recurrence and shows that observing part of the profile licenses a useful expectation about another part, or about how the system behaves under a relevant change.

4 NETWORK ORDER WITHOUT CONTROL

Khalidi's network account addresses a different shortcoming. A cluster is more than a list of associated properties, because its properties can be ordered, with some generating, sustaining, constraining, or explaining others. This is an advance over a bare cluster picture, since it explains why some properties are diagnostic and others merely derivative (Khalidi, 2013, 2018). Khalidi himself replaced the notion of a homeostatic mechanism with causal-network order in pursuit of a more general account, so identifying that order as one securing structure among several is continuous with his aim.

That order is the strength of the account. The present point is narrower: ordered dependence, even when objective and explanatory, does not by itself amount to feedback-sensitive correction.

Ordered dependence is still some distance from corrective control. A node in a causal network can support projection because the properties around it stand in systematic causal relations, and that is compatible with the system never tracking a departure, never allowing lower-scale variables to adjust, and never preserving a higher-scale relation through any such adjustment. Network order can therefore explain projectibility without amounting to homeostasis.

The second diagnostic accordingly asks whether the profile has internal structure, and the relevant evidence consists of asymmetries among the properties: which are upstream, which derivative, which central, and which only peripheral cues. The characteristic failure is the use of causal vocabulary for what is really an unordered list, since redescribing co-occurrence in causal terms does not by itself establish that any ordering obtains.

I use NETWORK ORDER broadly across the diagnostic. In strictly biological or mechanical cases the ordering will usually be causal, whereas in linguistic and social cases the relevant dependencies may instead be functional, conventional, sequential, or evidential. What matters in each case is that the profile has a structure that explains projection, because a merely associated bundle of features is not enough.

The broad use still needs a constraint. A non-causal ordering relation should be asymmetric: one feature conditions, licenses, constrains, or structures another. Evidential usefulness alone is not network order. A cue is evidentially privileged only when it tracks an ordering relation in the profile rather than merely co-occurring with the rest of the bundle.

5 FROM STABILIZERS TO CONTROLLERS

It helps to separate two questions that talk of maintenance tends to run together. Whatever keeps a cluster clustered can be called its STABILIZER, in the sense that removing it would let the category fragment, drift, or dissolve. The stabilizer is a functional role, and the causal structure that fills the role is the mechanism. This is the functional-role description of what Boyd called a homeostatic

mechanism (Boyd, 1991), and it has become a familiar way of talking about how linguistic and social categories are held together. Maintenance, in this sense, is the umbrella term.

The question the paper presses is which stabilizers are also controllers. A **CONTROLLER** is the narrower case: a stabilizer with feedback-sensitive corrective organization that preserves a higher-scale relation through lower-scale variation. Not every stabilizer meets this condition. Reliable co-occurrence can be maintained by entrenchment, by transmission, or by a frozen accident that nothing actively corrects, and ordered dependence can be maintained by causal wiring that carries no error signal and returns the system to no target.

The two ideas answer different counterfactuals, which is why they come apart. The stabilizer question is about the maintainer, and it asks what would happen to the cluster if the maintaining process were removed. The controller question is about the profile under threat, and it asks whether a perturbation to the profile is sensed and answered in a way that brings the profile back toward a relation worth projecting from. A cluster can answer the first question and fail the second, since entrenchment and transmission may hold a profile together without any of them detecting or correcting a departure. The space between a profile that is merely held in place and one that is actively corrected is where **HOMEOSTATIC** tends to be overclaimed.

Read this way, the three securing structures are three kinds of stabilizer. A stable cluster, a network-ordered kind, and a control-secured kind are all maintained, and they differ in how. Only the controller variety earns **HOMEOSTATIC** in anything like a literal sense, and the next section says what control adds, using the account that states the standard most carefully.

6 CORRECTIVE CONTROL

Control-secured kinds are more demanding, and the first thing to get right is the level at which the control is located. Kinds do not themselves sense, compare, or correct anything. A kind earns the label only when its projectible profile is maintained by a system or a practice that has feedback-sensitive corrective organization, so that the bearer of the control is the system rather than the category named.

Weinberger's central lesson is that homeostasis requires more than constancy, because a controlled system can preserve a higher-scale relation precisely by letting its lower-scale variables move. His dynamic causal models show that feedback systems, and the Watt governor in particular, produce control relations that static causal models and a simple vocabulary of intervention tend to obscure (Weinberger, 2026). Two consequences are worth drawing out. The first is that holding a variable rigidly fixed can be a sign of less control rather than more, since appropriate variation in the lower-scale variables is often the means by which the higher-scale relation is kept in place. The second is that not every intervention is equally probative. Perturbations to workload or demand can reveal the controller's response, while clamping the regulated variable itself can sever the very feedback that constitutes the control. Stillness under a clamp is therefore not evidence that nothing was being controlled.

For the purposes of this paper, corrective control has four evidential marks. There must be a perturbation that threatens the projectible profile; a variable or relation that the system tracks and that matters to the profile; a response pathway that changes lower-scale conditions; and a result in which that response preserves a higher-scale relation supporting projection. Feedback on its own does not satisfy these conditions, because what has to be shown is that the feedback explains why the profile remains projectible when it is perturbed.

The biological control literature makes the same warning concrete. Control mechanisms sense

Claim being made	Evidence needed	Failure mode
Stable cluster	Reliable co-occurrence across relevant contexts, plus a specified projection from observed to unobserved features	Mere recurrence without projectible payoff
Network-ordered kind	Internal ordering among properties, with some properties upstream, central, derivative, or evidentially privileged	Causal vocabulary used for an unordered list
Homeostatic-control-secured kind	Perturbation-sensitive correction that preserves a higher-scale relation through lower-scale variation	Treating stability, maintenance, or feedback as control

Table 1: Diagnostics for three ways projectibility can be secured.

conditions, transmit signals, and alter constraints within the mechanisms they control (Bich & Bechtel, 2022a), and in organisms these controls are integrated across the system rather than issued from a single command point (Bich & Bechtel, 2022b; Bich et al., 2016). This is a good deal more specific than the claim that a property cluster is stable, or that its properties stand in some causal order, and it is the third and most demanding of the diagnostics.

7 THREE DIAGNOSTICS

The proposal is best read as a diagnostic grid rather than as a single ladder. Control-secured cases will usually display stability and network order as well, but the inference does not run in reverse, since a stable cluster need not have an internally ordered causal profile and a network-ordered kind need not be maintained by any corrective control.

The grid is not an exhaustive taxonomy of every possible securing structure. It isolates three roles that are recurrently conflated in HPC discussions: reliable co-occurrence, ordered dependence, and corrective control. Selection, drift, institutional scaffolding, exemplar effects, and path dependence may each stabilize projectible profiles in ways that deserve separate treatment.

The grid is meant to turn the question around. For any proposed HPC kind, it asks three things before the word *HOMEOSTATIC* is used. The first is what the category lets us project. The second is what structure secures that projection, whether stable co-occurrence, network order, or corrective control. The third is what evidence would move the case from one diagnostic to another. The aim is not to police a word but to make the evidential demand behind it explicit.

Nothing in the argument requires every historical use of *HOMEOSTATIC* in HPC theory to have meant feedback control. The claim is normative and diagnostic: corrective control is the structure that gives the term distinctive content once stability, maintenance, and order have been separated.

8 HOW TO READ THE CASES

The cases that follow do not all have the same standing, and the differences are part of the point. The Watt governor is not put forward as a kind at all; it serves to calibrate the control structure that a homeostatic-control-secured kind would have to involve. Glycemic regulation is a candidate biolo-

gical domain in which that calibration can be applied, and in which the familiar set-point picture is itself under pressure. Interjections are a linguistic category whose stable projectibility is clear enough, and they are used to test whether stability and interactional order are sufficient for the stronger label, which they turn out not to be. Agreement monitoring is a candidate positive linguistic case, where the stronger label comes closest to being earned once the relevant repair loop is specified.

Running these cases across a machine, an organism, and two linguistic categories is deliberate. The diagnostic concerns how projectibility is secured, and that is not a question internal to any one science. If the same three questions discriminate among a governor, a regulatory profile, and two kinds of linguistic patterning, then the distinction is doing work in the theory of kinds rather than reporting the folklore of a particular field.

9 CALIBRATION CASE: WATT GOVERNOR

The Watt governor is a useful calibration case because its control structure is explicit. A change in engine speed moves the governor's flyballs; their position changes the valve aperture; the aperture changes the flow of steam; and the steam flow changes the speed of the shaft. The controlled target is shaft speed, mediated by the admission of steam, and a clamp placed directly on shaft speed would sever the incoming causal relations and destroy the control relation altogether (Weinberger, 2026).

That point does not make the governor immune to intervention. A change in workload or demand is the right kind of perturbation, because it lets the feedback relation operate and reveals whether lower-scale adjustment preserves the higher-scale coupling.

The governor shows clearly why HOMEOSTATIC ought not to mean constant. Its lower-scale variables are allowed to vary precisely so that workload and steam flow remain coupled in a stable higher-scale relation. This is the pattern the diagnostic is looking for, in which a perturbation is met by responsive adjustment in the lower-scale variables, and a relation that supports projection is preserved through that adjustment.

Because the governor is an artifact with a target fixed by its designer, it shows the control structure in an especially clean form, and that is what makes it a reference point rather than a kind. A candidate kind can then be tested against it, and the question for any proposed homeostatic kind becomes whether it exhibits this structure, rather than whether its properties happen to stay put.

10 BIOLOGICAL EXTENSION: GLYCEMIC REGULATION

Glycemic regulation is instructive because it is harder than the governor. The relevant candidate is not glucose as a substance, nor a single blood value, but the normoglycemic regulatory profile of an organism. The standard endocrine picture already supplies a control architecture, with a device that sets a normal range, a sensor that measures the regulated variable, a detector that compares the measured value against the range and emits an error signal, a controller, and an effector that moves the variable back toward the range (Bich et al., 2020). Taken at face value, this looks control-secured.

The projectible payoff is the organism's metabolic response across changing conditions. Given the normoglycemic regulatory profile, one expects coordinated changes in insulin, glucagon, hepatic output, and glucose uptake across feeding, fasting, and activity, rather than an isolated reading that happens to fall within a range.

It is worth noticing that the standard model already reserves the word CONTROLLER for the component that reads the error signal and drives the effector. The vocabulary of the calibration case and

of the biological case coincides, and not by accident, since both are feedback architectures in which a sensed departure drives a correcting response. The question is never whether the word fits the diagram, but whether the organism realizes the architecture that the diagram describes.

Two complications make that question sharper. The first concerns the set point, since population correlations for glucose and calcium may fit an equilibrium model better than they fit a defended set point (Fitzgerald & Bean, 2018). The second is that Bich et al. (2020) argue the feedback-loop diagram flattens a hierarchical organism into a chain of components, encourages a search for neatly localized parts, and assumes a normal range without explaining what establishes it. They move instead to organizational closure, on which glucose regulation is part of the organism's self-maintenance rather than the tracking of an externally given number. Read this way, the corrective structure is real, but the set point is the element the cybernetic analogy gets wrong.

That move marks the real contrast with the governor. The governor's target is set from outside by a designer, whereas the organism's range is constituted by what keeps the organism alive. The higher-scale relation that glycemic control preserves is therefore not a number but a relation between glucose availability and the organism's continued functioning, held in place through wide variation in the lower-scale variables across feeding, fasting, and activity. A biological example counts as homeostatic-control-secured only when the evidence displays this organized correction, and not when the variable merely stays within range. The question is whether departures are sensed and answered in a way that preserves the higher-scale relation, so that what the system returns to is best understood as the relation rather than the reading.

11 CONTRAST CASE: INTERJECTIONS

English interjections are the negative linguistic case. They look stable and projectible in use, since prosodic isolation, low syntactic integration, non-inflection, non-referentiality, and interactional force tend to travel together (Ameka, 1992; Norrick, 2009). Observing some of these properties licenses expectations about the others, which makes the profile projectible rather than a mere list.

The payoff is pragmatic. Forms such as *huh?*, *mmhm*, and *oh* support expectations about sequential position and interactional uptake, including repair initiation, continuation, and a change in epistemic state (Dingemanse, 2024), while forms such as *wow*, *oops*, *hey*, *well*, and *uh* invite expectations about stance, event management, attention, transition, or delay. Norrick (2009) is helpful here because he treats interjections as an open-ended class with recurrent pragmatic-marker functions.

The list is deliberately heterogeneous, since several of these forms are analysed as discourse markers, fillers, summonses, or response tokens rather than as central interjections. This heterogeneity is part of the test, because the contrast case is not a classical kind with settled boundaries but a projectible interactional profile whose kindhood is itself partly at issue.

The likely stabilizers include conventionalization, turn-taking routines, prosodic recognizability, acquisition, register, uptake, and metapragmatic learning, and together these make the category stable and useful. They may also give parts of the profile some network order, since sequential position, prosodic packaging, and syntactic non-integration are not wholly independent of one another.

The homeostasis question is more demanding. For interjections, the higher-scale relation would be an interactional profile linking turn position, prosodic packaging, low integration, and uptake. The lower-scale variables would include token form, intonation, sequential placement, phonological reduction, and local conventional meaning, and the relevant perturbations would include syntactic

integration, inflection, shifted prosody, misplaced sequential use, or failure of uptake. A homeostatic-control claim would require evidence that perturbations of this sort reliably trigger coordinated correction that preserves the higher-scale interactional relation. Repair, failures of uptake, teaching, and metalinguistic policing may sometimes furnish control-like evidence, but the evidence has to be produced, and until it is, interjections belong under the stable or network-ordered diagnostics. The case is useful precisely because it makes the temptation to overclaim visible and then declines to indulge it.

12 CANDIDATE POSITIVE CASE: AGREEMENT MONITORING

Agreement monitoring is the linguistic case where the stronger label comes closest to being earned. In agreement systems, one expression fixes features that other expressions must express. In Corbett's terms, the source is the controller, the agreeing expressions are targets, and canonical agreement is the case in which the features take matching rather than mismatching values (Corbett, 2006). The features may be number, person, gender, noun class, or case. Gender is therefore one possible feature in the system, not the object of the case.

That terminology is useful but not decisive. Corbett's controller-target relation identifies a dependency structure in agreement. By itself, it is not a feedback-control architecture. The controller does not sense a mismatch, compare target forms with a standard, or issue a corrective signal. Agreement morphology establishes the relation whose preservation matters; production monitoring and self-repair are where a control-theoretic claim has to look for feedback.

The projectible payoff is clear. Once the controller's features and the agreement domain are known, target forms can be projected across the domain. Number agreement lets us project verb form from subject number. Noun-class agreement lets us project class-marking across verbs, adjectives, and demonstratives (Ashton, 1947; Corbett, 2006). The dependency is also ordered rather than merely stable, because the controller-target relation is asymmetric: the controller fixes the relevant feature values, and targets express them.

The control question arises only when that dependency is embedded in a corrective process. Agreement-attraction errors show the perturbation clearly: speakers can be drawn toward a nearby mismatching noun and produce an agreement error (Bock & Miller, 1991). In online production monitoring, a speaker produces target forms, monitors the developing utterance, detects trouble, and repairs or recasts when the form fails to preserve the intended agreement relation (Levelt, 1983). On that loop, the tracked relation is agreement within the utterance, the lower-scale variables are particular target forms, and the response pathway is repair or recasting.

This keeps constraint and correction apart. A grammar may mark a mismatch as ill-formed without thereby supplying a corrective pathway. Corrective control requires the additional process by which a departure is detected and answered. The four marks may be met in the online repair loop: the mismatch threatens the projectible profile, agreement is the tracked relation, repair changes the lower-scale forms, and the higher-scale relation is preserved through that change.

Acquisition and normative correction are different candidate loops, not parts of a single system unless coordination is shown. In acquisition, learners converge on agreement patterns over developmental time; in normative contexts, teachers, editors, or interlocutors may reject or correct mismatches. These processes help stabilize agreement and may provide further control-like evidence, but they operate at different scales from online monitoring. The honest conclusion is conditional:

agreement monitoring is not trivially homeostatic, but it gives the paper a specified linguistic loop rather than a loose analogy from grammatical dependency to feedback control.

Grammaticality at large is a harder and more contested candidate. If correction, repair, acquisition, transmission, and uptake together formed an architecture that returned usage to a community-sensitive profile, grammaticality could turn out to be control-secured, but whether grammaticality is a single kind, and which loops would be doing the controlling, is just what is in dispute. The diagnostic leaves the possibility open without granting it, and agreement monitoring shows how a positive case would have to specify its loop.

13 WHAT THE DIAGNOSTIC CHANGES FOR HPC THEORY

The distinction matters for the wider HPC programme in linguistics and social ontology. Linguistic categories can be stable, projectible, and mechanism-maintained without all of them being homeostatic in the strict control-theoretic sense, and social practices can be stabilized by records, offices, sanctions, training, and uptake without thereby constituting single homeostatic control systems. The diagnostic asks each case to display its securing structure rather than inferring control from the mere fact of persistence.

A natural objection is that social correction and metalinguistic policing are themselves forms of feedback, and sometimes they are. The concession strengthens the diagnostic rather than weakening it, because it means that some linguistic and social categories may indeed be control-secured, but only where correction, repair, acquisition, transmission, and uptake combine into a coordinated architecture of the kind that agreement monitoring may exhibit. The point is that the evidence has to earn the upgrade case by case, rather than the label being granted on the strength of stability alone.

The distinction also bears on the more formal side of HPC work. It is tempting to operationalize homeostasis as a return-to-basin claim, asking whether a variable comes back to a set point after it has been perturbed. The governor and glycemic regulation both show why that test is too narrow, since what each system preserves is a relation, between workload and steam flow in the one case and between glucose availability and self-maintenance in the other, held in place through variation in the lower-scale variables. A test that measures only the return of a variable to a fixed value can miss control that works precisely by allowing the variable to move. The basin is better defined over the relation the system preserves than over any set of fixed values.

For HPC work in linguistics this implies a division of labour. Many of the categories the framework would call homeostatic are, on inspection, stabilized clusters or network-ordered kinds, which are real, projectible, and worth a realist's defence without being control-secured. Reserving HOMEOSTATIC for the cases that display corrective control leaves the others untouched, since they remain real and projectible, while showing the analyst what to look for, and what would count as having found it, before the strongest description is reached for. What philosophers of science gain is a domain-neutral test that distinguishes projectible stability, ordered dependence, and corrective control, and that says what evidence would move a case from one to another.

14 CONCLUSION

Projectibility remains the point throughout. Mechanisms and network order matter because they explain why projection is reliable, and homeostasis is one important way in which projection can be secured, but the word should be reserved for cases of corrective control. Drawn this way, the

diagnostic protects Boyd's anti-essentialist insight while keeping HOMEOSTATIC from becoming a loose synonym for stable, useful, or causally organized.

ACKNOWLEDGEMENTS

Drafting and revision benefited from OpenAI Codex (GPT-5), GPT-5.5 Pro, and Claude Opus 4.8. All content has been reviewed and revised by the author, who takes full responsibility for the final text.

REFERENCES

- Ameka, F. (1992). Interjections: The universal yet neglected part of speech. *Journal of Pragmatics*, 18(2–3), 101–118. [https://doi.org/10.1016/0378-2166\(92\)90048-G](https://doi.org/10.1016/0378-2166(92)90048-G)
- Ashton, E. O. (1947). *Swahili grammar (including intonation)*. Longmans, Green; Co.
- Bich, L., & Bechtel, W. (2022a). Control mechanisms: Explaining the integration and versatility of biological organisms. *Adaptive Behavior*, 30(5), 389–407. <https://doi.org/10.1177/10597123221074429>
- Bich, L., & Bechtel, W. (2022b). Organization needs organization: Understanding integrated control in living organisms. *Studies in History and Philosophy of Science*, 93, 96–106. <https://doi.org/10.1016/j.shpsa.2022.03.005>
- Bich, L., Mossio, M., Ruiz-Mirazo, K., & Moreno, A. (2016). Biological regulation: Controlling the system from within. *Biology & Philosophy*, 31(2), 237–265. <https://doi.org/10.1007/s10539-015-9497-8>
- Bich, L., Mossio, M., & Soto, A. M. (2020). Glycemia regulation: From feedback loops to organizational closure. *Frontiers in Physiology*, 11, Article 69. <https://doi.org/10.3389/fphys.2020.00069>
- Bock, K., & Miller, C. A. (1991). Broken agreement. *Cognitive Psychology*, 23(1), 45–93. [https://doi.org/10.1016/0010-0285\(91\)90003-7](https://doi.org/10.1016/0010-0285(91)90003-7)
- Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, 61(1–2), 127–148. <https://doi.org/10.1007/BF00385837>
- Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–186). MIT Press. <https://doi.org/10.7551/mitpress/6396.003.0012>
- Corbett, G. G. (2006). *Agreement*. Cambridge University Press.
- Dingemanse, M. (2024). Interjections at the heart of language. *Annual Review of Linguistics*, 10(1), 257–277. <https://doi.org/10.1146/annurev-linguistics-031422-124743>
- Fitzgerald, S. P., & Bean, N. G. (2018). Population correlations do not support the existence of set points for blood levels of calcium or glucose: A new model for homeostasis. *Physiological Reports*, 6(1), Article e13551. <https://doi.org/10.14814/phy2.13551>
- Goodman, N. (1983). *Fact, fiction, and forecast* (4th ed.). Harvard University Press. (Original work published 1955)
- Khalidi, M. A. (2013). *Natural categories and human kinds: Classification in the natural and social sciences*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511998553>
- Khalidi, M. A. (2018). Natural kinds as nodes in causal networks. *Synthese*, 195(4), 1379–1396. <https://doi.org/10.1007/s11229-015-0841-y>

- Levelt, W. J. M. (1983). Monitoring and self-repair in speech. *Cognition*, 14(1), 41–104. [https://doi.org/10.1016/0010-0277\(83\)90026-4](https://doi.org/10.1016/0010-0277(83)90026-4)
- Norrick, N. R. (2009). Interjections as pragmatic markers. *Journal of Pragmatics*, 41(5), 866–891. <https://doi.org/10.1016/j.pragma.2008.08.005>
- Onishi, Y., & Serpico, D. (2022). Homeostatic property cluster theory without homeostatic mechanisms: Two recent attempts and their costs. *Journal for General Philosophy of Science*, 53, 61–82. <https://doi.org/10.1007/s10838-020-09527-1>
- Slater, M. H. (2015). Natural kindness. *The British Journal for the Philosophy of Science*, 66(2), 375–411. <https://doi.org/10.1093/bjps/axt033>
- Weinberger, N. (2026). Homeostasis and causal control. *Biology & Philosophy*, 41(2), Article 18. <https://doi.org/10.1007/s10539-026-10018-8>
- Woodward, J. (2001). Law and explanation in Biology: Invariance is the kind of stability that matters. *Philosophy of Science*, 68(1), 1–20. <https://doi.org/10.1086/392863>